PROJECT TITLE	Building Australia's next-generation ocean-sea ice model

PROJECT QUALITY AND INNOVATION

Introduction: The oceans and sea ice are critical components of Earth's climate and weather, acting to modify the rate of terrestrial climate change, controlling low-frequency climate variability, governing global and regional sea level and influencing shorter-timescale weather events. Our sole means of predicting the future state of the ocean and sea ice is through numerical models, founded on physical laws and running on high performance computers. These models form the basis of global climate projections, are used for ocean state forecasts and reanalysis, and in the coming years will underpin sea ice forecasts, surface wave forecasts and coupled numerical weather prediction. **This project will create a world class, next-generation ocean-sea ice model for Australian users, combining the latest international developments with Australian expertise.** The model will directly contribute to improved future climate projections and decadal forecasts, and to maritime operations (e.g. search and rescue, defence and shipping) around Australia and near Antarctica. The model will also be a valuable and widely-used research tool, used to better understand the dynamics of the ocean-wave-sea ice system, and will contribute to Australia's preparations for impending climate change by providing a platform for coupled ocean-atmosphere-land-sea ice simulations.

Ocean-sea ice model development involves three main elements. First, the **codebase**, which requires enormous development, usually via international collaborations. Second, a **configuration**, tailored to local requirements, which involves creating ancillary files to drive the model and optimise for local high-performance computers. And third, the **computational infrastructure** which is needed to run the model and store/analyse the output. For high-resolution models, which are the most accurate and most valuable, the resources required to configure, run and analyse are now beyond the capacity of a single team or research organisation. This proposal outlines a nationally cooperative and collaborative approach to develop models that will enable the community to break new ground in marine forecasting and research.

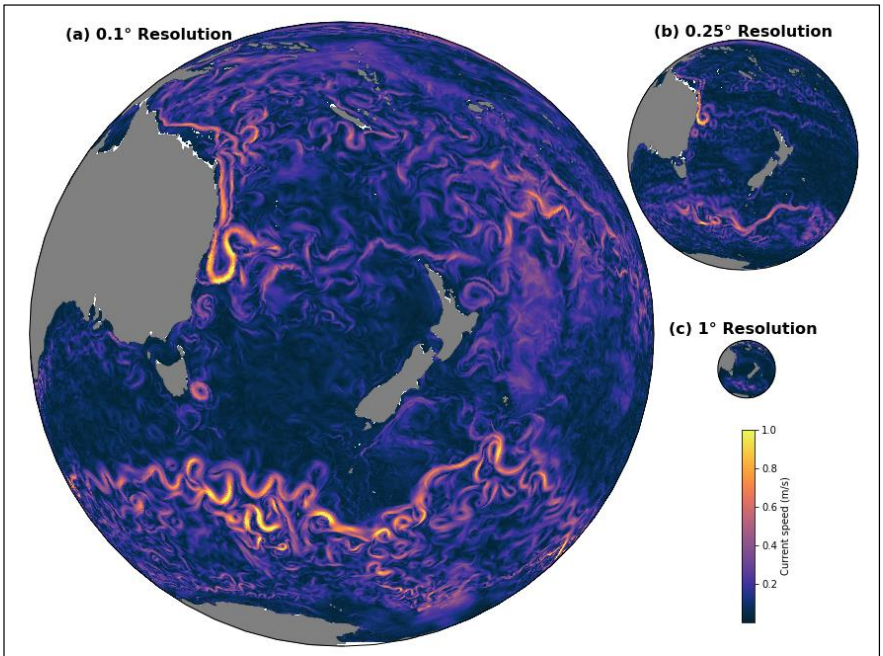

Figure 1: Snapshot of near-surface current speed in ACCESS-OM2 for (a) 0.1°; (b) 0.25°; and (c) 1° resolution. The size of each panel is scaled by its resolution to highlight differences in compute requirements and the physical representation of the circulation.

The Consortium for Ocean-Sea Ice Modelling in Australia (COSIMA) was formed to support the development of model configurations for the Australian community, with a focus on Southern Ocean science. COSIMA grew from foundational efforts to establish national partnerships that began in 2012, and received funding from an ARC Linkage Project in 2016 (LP160100073) which covers the salary of a research fellow and a part-time software engineer. From this modest funding base, COSIMA has transformed ocean-sea ice modelling in Australia, leveraging a community of >60 researchers to produce and freely distribute the ACCESS-OM2 model at 3 different resolutions (Kiss et al. 2020 and Fig. 1). There are currently >30 active research projects using ACCESS-OM2, and the model is now being adopted nationwide by (1) the Bureau of Meteorology (BoM), (2) the Australian Antarctic Division (AAD) and (3) CSIRO. **We are seeking to extend this extraordinary return on ARC and partner investment by proposing a new initiative to bring next-generation ocean-wave-sea ice models to the Australian community.**

Aims: The aims of this proposal are to:

1) Configure, evaluate and publish the next-generation MOM6 ocean model and CICE6 sea ice model, culminating in a new, world-class Australian ocean-sea ice model: "ACCESS-OM3";
2) Advance Australian capacity to model the ocean's biogeochemical cycles and surface waves, including the feedback between waves, sea ice, biogeochemistry and ocean circulation; and
3) Build on the success of COSIMA to establish deep ties between Australia's leading ocean-sea ice modelling institutions, while maintaining ACCESS-OM2 for ongoing research projects and operational products.

ACCESS-OM3 will be configured at a range of resolutions, with both regional and global implementations, to underpin ocean, sea ice and wave forecasts, future climate projections, decadal climate prediction and fundamental research into ocean, sea ice and biogeochemical processes.

Background:

Ocean-sea ice models sit at the nexus of physical oceanography, software engineering and high-performance computing. Fluid mechanics describes the physical laws which underpin these models, but the huge range of turbulent scales in the system make it infeasible to exactly solve these equations. Instead, approximate numerical solutions are required. These approximations imply a trade-off: fine spatial resolution (i.e. small grid boxes) allows the solution to remain more faithful to the physics, but requires more grid boxes over a given spatial domain, and the computational cost of this fine resolution is enormous. On the other hand, lower resolution simulations run faster and use less CPU time, but are less accurate and necessarily omit explicit representation of some physical processes (see Fig. 1 for an indication of the differences in scale and physical representation between high and low resolution). The acceptable degree of compromise depends upon the intended model purpose and computational constraints.

The primary task involved in building an ocean-sea ice model is to couple together the sea ice and ocean sub-models; in the case of ACCESS-OM2 (see Fig. 2a) this coupling between CICE5 (Hunke et al. 2013) and MOM5 (Griffies, 2012) was achieved with the OASIS library (Valcke et al. 2013). The coupler acts to interpolate fields and to pass fluxes (of heat, freshwater and momentum) between the model components, while also ensuring conservation. The coupler is also needed to interpolate atmospheric forcing fluxes (in ACCESS-OM2, these are from the JRA55-do dataset; Tsujino et al. 2018). The different timescales and computational schemes of the sub-models adds complexity to this coupling process. In particular, the interplay between sub-models affects model stability, optimisation and parameterisations within each component that may create model biases. This complexity is outlined in detail by Kiss et al. (2020), which forms the basis for future scientific and operational deployments of the ACCESS-OM2 model.

Model Outline

The performance of the ACCESS-OM2 model is on par with current international efforts to configure global ocean-sea ice models (Barnier et al. 2014; Hewitt et al. 2016). However, continued development is needed to keep Australian ocean modelling at the forefront of global developments. The present proposal will lead and coordinate that development, with a focus on bringing new tools and a world-leading suite of sub-models to the Australian community, culminating in the release of ACCESS-OM3. The new model will comprise:

MOM6: Our community has been well-served by the previous Modular Ocean Model v5 (MOM5), but it is now a legacy code that will no longer be supported or actively developed. Australia needs to adopt a new-generation system for all ocean modelling applications. The logical choice for this new model is MOM6 (Adcroft et al. 2019). Despite the similarity in name, MOM6 is a complete re-architecture of the model dynamical core, but retains the same external hooks as MOM5 to allow similar coupling to other model components. MOM6 is a huge step forward in technology over MOM5 (or other currently available global models) in that it uses a flexible Arbitrary Lagrangian Eulerian vertical coordinate (ALE; White et al. 2009), which allows for more configurable domains while minimising model errors through spurious mixing (Gibson et al. 2017). The COSIMA group has already been using MOM6 for idealised and test configurations over the last 5 years, and is well positioned to lead Australia into adoption of this innovative new code. MOM6 is now implemented as the ocean component of the coupled climate model developed by our partners at the National Ocean and Atmospheric Administration's (NOAA's) Geophysical Fluid Dynamics Laboratory (GFDL) and is ready for use by external collaborators.

The adoption of MOM6 will create a step change in the ocean modelling and ocean model development community in Australia. Specifically: a) its versatile vertical coordinate system can be configured to create a hybrid coordinate, which is tailored for individual regions; b) its wetting/drying capabilities will allow this model to extend into shallow coastal regions, and to include the modelling of oceanic cavities under marine-terminating glaciers on the Antarctic shelf; c) it includes a scheme to incorporate tides, which are essential for regional and high-resolution models, particularly in shallow regions and in ice-shelf cavities; and d) its ability to take long tracer timesteps greatly reduces the cost of incorporating a biogeochemical module into the model. The model's capability ranges from ultra-high-resolution regional configurations to global climate model simulations, leading to the possibility that this model could span the full range of ocean model applications used in Australia. Furthermore, our model improvements will contribute to, and be supported by, the open-source development effort into this new model, which is an international community already including over 40 GitHub contributors.

CICE6: The Community Ice CodE, CICE, is the established world-leader in sea-ice modelling. The ACCESS models currently use CICE5, but this version is no longer supported and there are clear advantages to upgrading to the latest version, CICE6. In CICE6, the code has been modularised to separate the vertical thermodynamics (known as Icepack) from the ice dynamics and rheology. CICE6 can also be built in stand-alone mode (a key advantage for code testing and optimisation) and has advanced biogeochemistry and fast-ice physics. Most critically, CICE6

includes ice-thickness and floe-size distributions (Horvat & Tziperman 2015; Roach et al. 2018) which are key elements in understanding the impact of surface waves on sea ice in the marginal ice zone. Finally, the upgrade to CICE6 will ease coupling with MOM6 (with CICE6 currently being configured to also run on the horizontal model C-grid used by MOM6, a significant change from MOM5). Therefore, the second major new initiative in this proposal will be to incorporate the latest developments in the CICE6 sea-ice model into our model framework.

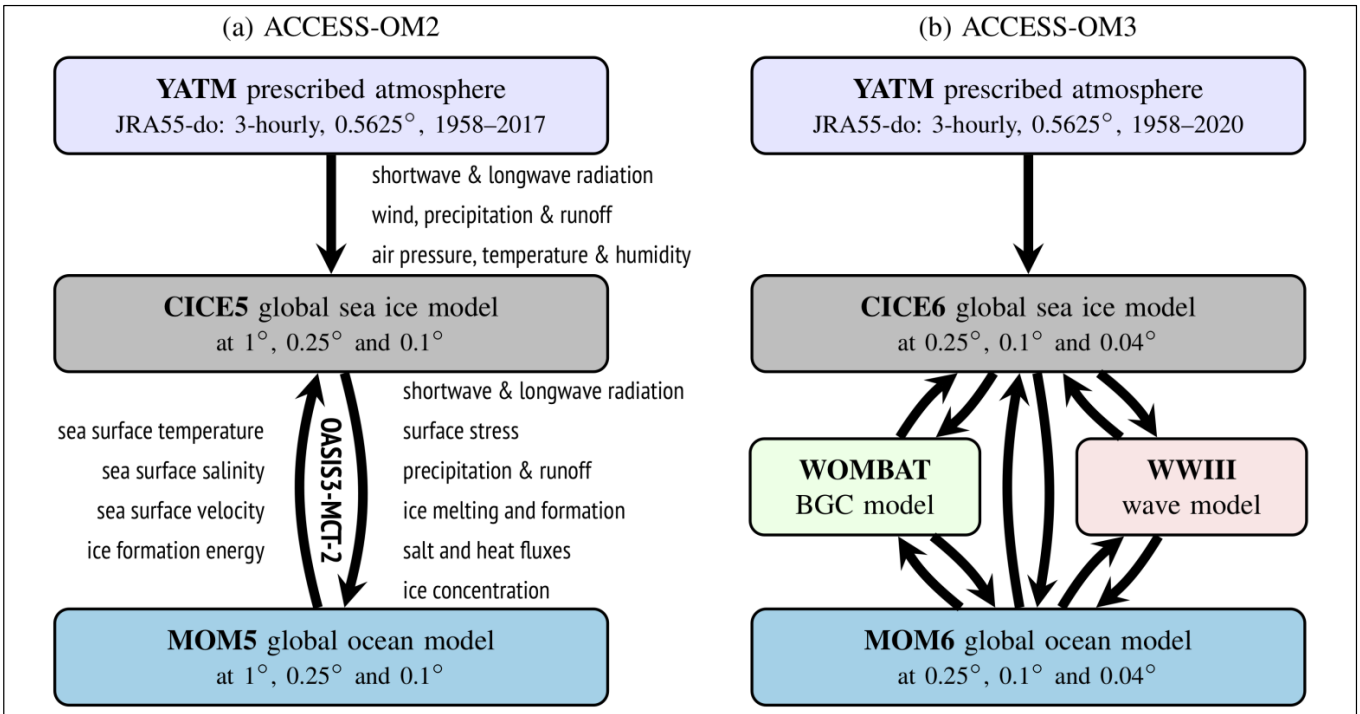

Figure 2: Schematic representation of coupled components of (a) the existing ACCESS-OM2 model (Kiss et al. 2020); and (b) the planned configuration for the future ACCESS-OM3 model.

WOMBAT: Inclusion of biogeochemical (BGC) modules into ocean and climate models has the twin benefits of prediction of important ecological variables, while BGC tracers provide additional metrics to evaluate model skill. Australian BGC modelling efforts with MOM5 use the World Ocean Model with Biogeochemistry And Trophic dynamics (WOMBAT; Oke et al. 2013). WOMBAT has been previously coupled into the Ocean Forecasting Australia Model (OFAM3) to investigate anthropogenic carbon uptake (Langlais et al, 2017, Menviel et al., 2018) and into the ACCESS Earth System Model (ACCESS-ESM1.5) for climate change projections with carbon-climate feedbacks (Ziehn et al., in review). However, use of this model is limited by the computational expense of advecting numerous biogeochemical tracers at high resolution. Incorporating WOMBAT into ACCESS-OM3 will have significant benefits; in particular, MOM6 allows us to take longer timesteps for passive tracers, reducing the computational cost of WOMBAT by a factor of 10-20. Embedding WOMBAT into ACCESS-OM3 will facilitate research into regional and global questions around water quality, ocean acidification, marine productivity, and oceanic uptake and storage of anthropogenic carbon. In addition, WOMBAT combined with BGC in CICE6 will enable the interaction between sea ice and ocean BGC for the first time in an Australian model.

WAVEWATCH-III: Many large-scale oceanic processes are coupled with surface waves, including weather, tropical cyclones, sea ice cover and upper-ocean mixing and turbulence. Inclusion of wave-ocean coupling also leads to reduction in biases of the simulated climate, including mixed layer depth (Walsh et al., 2017), and the addition of wave-induced mixing brings widespread improvement in sea surface temperature in both hemispheres in summer and winter (Stoney et al., 2018). Wave-induced mixing also leads to a reduction of seasonal extremes of sea ice extent, bringing the seasonal amplitude of sea ice extent significantly closer to observations, as well as causing an increase in sea ice thickness in Antarctic spring and summer, i.e. removing the well-known summer bias in climate models (Thomas et al., 2019). For these reasons, ACCESS-OM3 will optionally couple with the advanced WAVEWATCH-III model (WW3; see technical publication authored by WW3DG).

The inclusion of WW3 into the coupled ocean-sea ice model will allow COSIMA researchers to incorporate wave information into the surface boundary layer scheme (e.g. Babanin, 2006; Li et al. 2019), thereby reducing climate biases. It will also resolve nonlinear interactions between the wave-field and surface stresses, and interactions with sea ice, including the seasonal ice breakup. Thus, there is a clear imperative for coupled ocean-ice-wave modelling (Boutin et al. 2020) for the purposes of sea ice prediction, as well as for enhanced surface wave forecasts.

To summarise, this project will integrate the world-leading MOM6 ocean model with the new CICE6 sea ice model as well as biogeochemical and surface wave models (WOMBAT and WW3) to support the development of the new ACCESS-OM3 global ocean-sea ice model (Fig. 2b). The model will be configured at 3 different global resolutions, with the finest resolution grid spacing being 0.04° (a factor of 2.5 finer than the highest resolution currently available in Australia) along with flexibly deployable higher resolution regional configurations. ACCESS-OM3 will thereby provide a foundation for Australia's future ocean state forecasts, wave forecasts, sea ice forecasts, decadal climate prediction and biogeochemical predictions; and will form the first Work Package of the project. We will also develop sophisticated tools to interrogate, analyse and optimise our model configurations (Work Package II), and support the COSIMA community and legacy codes, such as ACCESS-OM2 (Work Package III). The detailed plan to achieve the goals of these three Work Packages is outlined in further detail below.

Work Package I: Creation of ACCESS-OM3

Over the last twenty years, large-scale ocean modelling in Australia has been powered by the Modular Ocean Model (MOM), created by our partners at the National Oceanic and Atmospheric Administration (NOAA) Geophysical Fluid Dynamics Laboratory (GFDL). This model underpins our national ocean forecasting system (Bluelink; BoM), our national ocean reanalysis system (BRAN; CSIRO), our coupled climate model and Earth system model (currently known as ACCESS-CM2 and ACCESS-ESM1.5, respectively) and the CAFÉ decadal prediction model under development at CSIRO. It is for this reason that the initial COSIMA model development effort has focussed on MOM5, ensuring that all of the above applications use a consistent framework, and are supported by an active and involved community with detailed knowledge of the component models.

The development of ACCESS-OM3 is designed to build on the success of ACCESS-OM2, while advancing Australia's position at the forefront of international ocean and sea ice modelling. This development will require coupling each of the model components listed above (see Fig. 2b), which will involve significant investment in software engineering and model testing. This task will form the basis of Work Package I, which will be the primary focus of the COSIMA group over the next 4 years and will demand priority resources and effort of the project. It will be a critical step in transforming Australia's capacity in modelling the ocean-sea ice system, and in supporting national operational needs. These development tasks require careful sequencing to optimise the development process, and to ensure growing scientific outputs will occur at every stage. This sequencing, and the scientific output associated with each development stage, is outlined in Fig. 3, with numbered reference to the following topics:

1) The first phase of the project will involve the adoption of a regional, stand-alone implementation of MOM6, led by CI Shakespeare. This model will have open boundaries that are driven by existing ACCESS-OM2 simulations, and use the JRA55-do atmospheric forcing that is used in global models (Tsujino et al. 2018). This regional implementation will be relocatable and will be scaled to progressively higher horizontal and vertical resolution – with the finest anticipated grid being a 500m (1/200°) resolution, 200-layer configuration that will allow us to resolve submesoscale dynamics and the generation of internal waves (e.g., Shakespeare & Hogg, 2017). Relevant domains of interest include the Tasman Sea, the Antarctic continental slope, the open Southern Ocean where eddies interact with bathymetry and the Indonesian throughflow. MOM6 will permit us to explicitly model tides in these regions, and we will test the wetting and drying capability in shallow coastal regions, with a view to enhancing modelling links between the open ocean and coastal ocean modelling communities in Australia. The regional implementation is a stepping-stone that allows us to test the viability and performance of the model at a range of model resolutions within computational constraints that will inform the architecture of the global model. However, it will also have direct benefit to ongoing process studies of small-scale features in the ocean, such as topographic interaction, internal wave generation and ocean mixing (e.g., Shakespeare & Hogg, 2019).

2) In Year 1, we will also begin work on adopting a stand-alone version of the new CICE6 sea ice model, led by PI Heil and CI Bennetts, both of whom have direct experience in the implementation of CICE. This stand-alone implementation is designed as a pathway to translate recent sea ice optimisation studies using ACCESS-OM2 to CICE6, to test scaling of the new model, and to better validate the new physics available in CICE6 – particularly the model's ability to carry and evolve the sea ice floe size distribution (Roach et al. 2018). The floe size distribution is the key innovation that will allow for the effects of surface waves on the dynamic and thermodynamic properties of the marginal ice zone, including breakup of consolidated sea ice in early summer and driving rapid advances of pancake ice cover in the winter. In addition, it will be prudent to investigate new code representing near-coastal processes, such as fast-ice evolution and polynya formation, in the context of the atmospheric forcing applied.

3) During Year 2, we will begin work on the global coupled MOM6-CICE6 model (Fig. 2b) that will form the basis of ACCESS-OM3. This program will be coordinated by the COSIMA Research Fellow, but also supported by all investigators and the Research Software Engineer employed on the project. It will rely heavily on our experience of building ACCESS-OM2, but differences in the horizontal and vertical grids, in the model architecture as well as

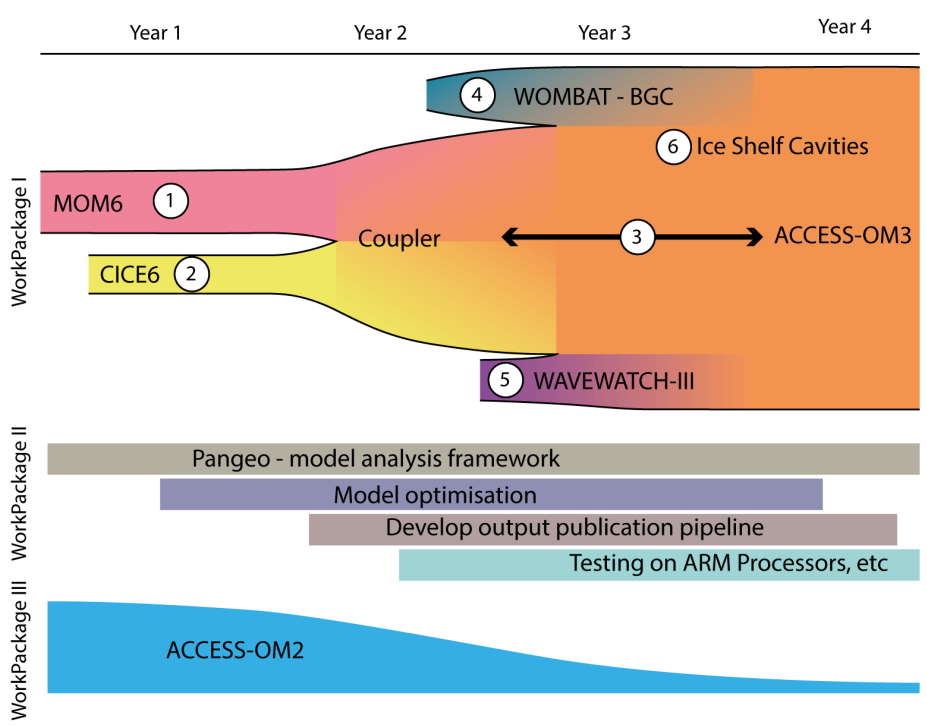

Figure 3: Outline of the project science plan, divided into 3 Work Packages. Work Package I is the largest, and requires careful sequencing – numbers in the diagram correspond to items listed in the text. Work Package II focusses on the computational and infrastructure side of the project. Work Package III focusses on maintaining ACCESS-OM2 for continuity of research and operational needs.

physics and in the interactions between each of these components will require significant development and careful testing. The model build will initially be undertaken at 0.25° resolution, the lowest resolution to be supported, and will be pushed to higher resolution (0.1° and 0.04°) once the initial coupling has been verified. The final product will be amongst the world's highest resolution global ocean-sea ice simulations.

Following the initial model build, three science teams will work on the processes that govern model biases in the new configurations (compared against *in situ* and satellite observations), and will endeavour to find methods to eliminate or minimise these biases. The first of these three teams, a tropical biases group (led by CI Holmes) will look at the stratification and circulation in the upper tropical ocean which is critical for global climate variability and decadal prediction. The sea-ice forecasting group, led by PI Sandery, PI Heil and CI Hobbs, will use advances made in optimising ACCESS-OM2 to compare the ACCESS-OM3 model state with observational products, with a particular emphasis on understanding errors in the sea-ice model and drivers of its predictability. Thirdly, a large-scale circulation group (led by CI England and PI Marsland) will evaluate circulation metrics such as the overturning circulation, model stratification, and transport by major current systems. The end product will be a major community-authored "model release" paper in the style of Kiss et al. (2020) which will document the model development, the benefits and biases associated with each resolution and recommendations for future improvements.

4) An important component of ocean model circulation studies is the inclusion of biogeochemical (BGC) models. Inclusion of BGC tracers provides additional constraints on the model physics and allows prediction of ecosystem response to the ocean circulation, and therefore is critical for potential end-users of our models, including fisheries and ecosystems researchers. The COSIMA team is currently working to integrate the WOMBAT biogeochemical model into ACCESS-OM2, and is expecting first results in the current calendar year. This model will be provided to the Ocean Model Intercomparison Project experiment database for analysis by international teams of researchers. However, ACCESS-OM2-WOMBAT is limited by computational resources – BGC models carry a number of tracers which, in the case of WOMBAT, can act to triple the required computational cost of the simulation. It is therefore not practical to include BGC at the highest resolution. The numerical scheme in MOM6 will eliminate this constraint, as it allows passive tracers (i.e. BGC tracers) to be advected at timesteps approximately 10-20 times longer than the dynamical timestep, without model instability (Adcroft et al. 2019). This advance will reduce the computational constraint of using WOMBAT by a factor of 10-20, and makes it practical to incorporate BGC at any resolution. The implementation of this model will be led by PI Matear (the primary developer of WOMBAT); he will also lead the team to evaluate the model. It is here that the stand-alone MOM6 model (see item 1 above) will be critical to the development sequence; in Year 2 we will incorporate WOMBAT into MOM6, and in Year 3 WOMBAT will be sequenced into the fully coupled ACCESS-OM3 model. The BGC component of ACCESS-OM3 will be further developed to support studies of BGC interaction between ocean and sea ice (a BGC module is built into CICE6), giving new capability to the Australian community.

5) Large-scale ocean and climate models cannot resolve surface waves on the ocean. Yet, evidence suggests that waves modulate the input of momentum to the ocean from wind forcing, mediate gas transfer and increase near-surface ocean mixing (Thomas et al. 2019). As models approach higher resolution, the effect of waves becomes more important. We will incorporate the world-leading wave model, WAVEWATCH-III (WW3), into our coupled model. Beginning in Year 3, a team led by CI Babanin (a member of the WW3 Development Group and responsible for some recent developments in WW3) will work on testing WW3 in the stand-alone ocean model. PI Brassington and his team at the BoM will also play a key role in this program. Computational inefficiency and poor scaling means that the wave model will be restricted to running at the resolution of the atmospheric forcing (~0.5°), so that the coupler will be needed to interpolate the wave model. This task therefore relies on completion of the coupled ocean-sea ice model implementation. In Year 4, this team will work on understanding the role of waves in the open ocean, which will be combined with the advances from Task 2 above to study the effect of surface waves in the marginal ice zone, in a coupled ocean-ice-wave framework. Once implemented, we will use this model to study (i) rapid advances/retreats of pancake ice during winter storms, driven by a combination of waves, winds and the ocean, and (ii) ocean-ice-atmosphere feedbacks resulting from wave-induced breakup of consolidated ice cover that affects momentum and heat fluxes by, for example, changing the boundary-layer roughness of the ice cover and ice melt rates (Bennetts et al. 2017).

6) One of the major uncertainties in future sea level rise is the role of melting Antarctic glaciers in adding fresh water to the ocean. This process depends upon the rate of melting of ice shelves from underneath, by the ocean; but data in these regions is very sparse, and few models are able to simulate these under-ice dynamics. The innovation of the ALE vertical coordinate in MOM6 will allow us to model these under-ice-shelf cavities, including the contribution of tides. In Years 3 and 4 a team led by CIs Morrison and Spence will use the MOM6 stand-alone model to test and calibrate these dynamics in key regions of the Antarctic coastline such as the Ross Sea and Bellingshausen Sea coastline. This technical development will supplement existing regional-scale projects at the AAD to couple ocean models with Antarctic ice sheet models (led by Dr Galton-Fenzi, listed as an AAD contribution to the project).

ACCESS-OM3 will take the first three years of this project to mature. This development will be an open project – all code development will be conducted transparently on the COSIMA GitHub repository and be available for the community to use at any time. The COSIMA team will support model users to download and run the model, and outputs from all of our runs will be available on the COSIMA data project that is available for all users to join. The model will benefit from wider community testing and engagement; it will then be adopted for operational purposes by our end-users and partners, such as Bluelink (including the Department of Defence), the CAFÉ decadal prediction study and researchers in the university community and the ARC Centre of Excellence for Climate Extremes.

Work Package II: Developing an optimisation and analysis pipeline

Over the last 4 years of the COSIMA project, our team has learnt to appreciate that the most significant constraints on both model development and scientific research emerge from the computational infrastructure. In short, high-resolution models are limited by their scalable performance and throughput, the flexibility of the code to transition to utilise new High-Performance Computing (HPC) hardware (including global trends in HPC hardware which are beyond the control of science communities), by our capacity to store and publish the model output, and by the innovation in software that can better analyse that output. As just one example of the software transition, we have recently explored how to analyse output from the 0.1° version of ACCESS-OM2 with a new parallelised software stack using dask and xarray at NCI. These technologies did not exist when COSIMA was initially formed; we have adopted these innovations through a home-made tool known as the COSIMA Cookbook (the subject of a recent community hackathon with 25 participants). Furthermore, this software is better suited to scaling beyond the current 48 cores currently used. The changes in software and analysis platforms is moving so rapidly that a focussed Work Package is needed to keep pace.

This Work Package, to be led by CIs Hogg and Evans, will incorporate the following tasks (as sequenced in Fig. 3): (1) Further development of a model output analysis environment based on the Pangeo initiative and the COSIMA Cookbook (all Years); (2) Optimisation of each model component, independently (Year 1) and together (Years 2 and 3), in collaboration with NCI's HPC scaling and optimisation team; (3) Strengthening our pipeline for publishing model output (see the COSIMA model output collection hosted by NCI at this doi:10.4225/41/5a2dc8543105a); and (4) Testing our codes on next-generation compute architecture, including Fujitsu's ARM processors (to be used in Japan's upcoming first exascale HPC system "Fugaku") to future-proof our model developments (Years 3-4).

Work Package III: Maintaining ACCESS-OM2

The huge strides taken during the last 4 years of COSIMA funding have been focussed on creating, configuring, optimising and documenting the ACCESS-OM2 model. This world-class model can be run at 3 resolutions, up to the state-of-the-art 0.1° configuration, which runs on 5184 cores on NCI's new Gadi supercomputer and consumes

50,000 core hours per year of simulation. This model has already supported a wide range of research being undertaken at universities and publicly-funded research institutions, including research into marine heatwaves (Pilo et al. 2019), ocean heat and momentum uptake (Stewart & Hogg, 2019), ocean transport mechanisms (Boeira Dias et al. 2020), Antarctic ice sheet interactions (Moorman et al., in review) and international model intercomparisons (Hirschi et al. 2020). Current projects in Lagrangian methods, Southern Ocean mode water formation, El Niño, sea ice sensitivity, mesoscale eddy detection, the Indonesian throughflow and biogeochemistry underscore the critical role played by the model infrastructure. This model also underpins the next version of the national operational ocean forecasting system, Bluelink, the ACCESS-CM2 coupled climate model and the ACCESS-ESMv1.5 Earth System Model. Maintenance of this software infrastructure is critical for the ocean and sea ice community in Australia.

Software infrastructure such as ACCESS-OM2 has a typical lifetime that exceeds 5 years. For example, many new PhD students just starting their projects employ this model, and will do so until the next generation of code becomes available. Our community continues to publish results from model simulations conducted 5 years ago, even before the formation of COSIMA. Moreover, Bluelink forward planning highlights that ACCESS-OM2 will be required for operational forecasts until at least 2023. It follows that continued effort in supporting ACCESS-OM2 is critical, even though newer tools are in the development pipeline. Work Package III will undertake this effort, including: (i) testing and releasing code that is compatible with upgraded hardware and compilers; (ii) maintenance of the open-source code repository with expected bug fixes and configuration improvements; and (iii) supporting new model users.

The main goal of this Work Package is therefore to continue supporting the collaborative community that we have built over the past 4 years. The resources expended on this task will decay with time over the life of the 4-year project (see Fig. 3) but the value of the model for end-user applications and scientific research will continue undiminished.

INVESTIGATOR(S)/CAPABILITY

This is a broad collaboration between university researchers and scientists in partner institutions that covers a wide range of model development activities and scientific end users. The team is correspondingly large, with 11 Chief Investigators and 7 Partner Investigators. The team will be led by Professor A. McC. Hogg, who leads the current COSIMA project. Professor Hogg has extensive experience in high-resolution ocean modelling and in the analysis of mesoscale dynamics and ocean processes. He has driven the formation of the COSIMA consortium from inception. Prof Hogg will supervise the COSIMA Research Fellow and Research Software Engineer employed by the project, and will take responsibility for coordination and communication of all COSIMA activities including weekly video meetings and annual workshops; he will devote 0.2 FTE to the project. He will be ably supported in this task by a large team of CIs and PIs with a wealth of experience across the disciplines encompassed by this project. The contribution of each investigator is outlined below.

Professor M. H. England has a sustained successful track record of leading large projects (3 x CoE's, ARC Federation Fellowship 2006-2010; ARC Laureate Fellowship 2011-2015). Prof England will dedicate 0.1 FTE to delivering the outcomes of this project; he will support Prof Hogg to lead the project, will have an oversight role in the development of ACCESS-OM3 and will co-lead the evaluation of large-scale circulation biases in the new model. Such strong engagement from one of Australia's leading ocean scientists indicates his commitment to this vital collaboration.

Dr A. K. Morrison is a DECRA fellow with specific expertise in the processes driving and modifying Southern Ocean circulation. She has extensive experience using both global and idealised eddy-rich ocean models to investigate Southern Ocean dynamics. She will contribute 0.1 FTE to this project, and will engage with the MOM6 regional model, while co-leading the ice-shelf cavity experiments.

Dr P. Spence is an ARC Future Fellow who has played a critical role in motivating and developing high-resolution ocean-sea ice models, and was primarily responsible for bringing the MOM-SIS-025 model (which pre-dated ACCESS-OM2) to Australia. Dr Spence (0.1 FTE) will contribute his broad experience in model development to the ACCESS-OM3 development process, and will co-lead the ice-shelf cavity experiments with Dr Morrison.

A/Prof L. Bennetts is an ARC Future Fellow who will bring his experience in advances in marginal ice zone modelling to the COSIMA community, contributing 0.1 FTE to the project. In particular, the latest release of the CICE sea ice model (v6.1) contains prognostic floe size information and wave-energy distributions in the marginal ice zone (initiated by Dr Bennetts' DECRA, DE130101571). He will work with PI Heil to establish the CICE6 stand-alone model, and will work to include recent advances in waves-in-ice modelling.

Professor A. Babanin is an expert in surface waves and their representation in models. His contribution to the project (0.1 FTE) will be coupling waves (the WAVEWATCH-III model) into ACCESS-OM3, including wave-ocean coupled effects (wave-induced turbulent mixing, Stokes drift) in the upper ocean layers and coupled wave-sea ice interaction in polar ocean regions (Arctic and Antarctic).

Dr Ryan Holmes is an expert in tropical ocean dynamics, ocean mixing and water-mass transformation. He will contribute 0.1 FTE to the project, in two primary areas. Firstly, he will lead the assessment and development of the representation of the tropical oceans by ACCESS-OM3 across resolutions. He will focus particularly on optimizing the parameterisation of vertical turbulent mixing in the upper equatorial ocean to improve the model representation of upper ocean heat variability on seasonal to decadal time-scales. Secondly, he will lead an effort to implement and assess water-mass transformation diagnostics budgets using new techniques (e.g. Holmes et al. 2019).

Dr Ben Evans is the Deputy Director of the National Computation Infrastructure (NCI) at ANU and will dedicate 0.1 FTE to the project. Dr Evans' main role will be to co-lead Work Package II with Prof Hogg. Dr Evans and his team will improve the performance and porting of the code to the new Gadi High Performance Computing system at NCI. He will lead the investigation of the portability and performance of the code to the next generation of hardware architectures and associated software stacks such as the follow-on to the current NCI system, including new ARM processor technologies currently emerging in Fujitsu and Cray systems.

Dr W. Hobbs is employed full-time as a sea ice-ocean interaction researcher at the Australian Antarctic Program Partnership. His will commit 0.1 FTE to the project, focussed on the polar Southern Ocean, and in particular he will co-lead the sea ice bias evaluation in ACCESS-OM3 (with Dr Sandery and Dr Heil) to analyse how processes at the ocean-sea ice interface impact the model's representation of both sea ice and Southern Ocean water masses.

Dr C. J. Shakespeare is a DECRA Fellow who will contribute 0.1 FTE to this proposal by leading the program to develop regional model configurations in MOM6 to investigate internal wave and other small-scale dynamics, building on his current experience running similar regional models with MITgcm. Dr Shakespeare will lead the development of new analysis methodologies for such models and his team will also utilise the regional MOM6 configurations to contribute to the development of parameterisations for internal wave processes.

A/Prof A. Toffoli has been active in modelling wave–ice interactions since 2013. He will contribute 0.1 FTE to the COSIMA community by developing new observation-based physics describing sea ice dynamics and atmosphere-sea ice-ocean interaction. In collaboration with Dr Bennetts, he will implement and validate advances in marginal ice zone physics in a coupled model framework.

Dr G. Brassington leads the ocean forecasting team in the Bureau of Meteorology. In addition to providing overall management for the project, he will directly supervise BoM staff (Dr J. Freeman, Dr S. Zeiger, Dr P. Sakov, Dr P. Divakaran and Dr F. Colberg) responsible for the technical contributions from the BoM side (totalling 0.5 FTE). He will contribute 0.05 FTE to the project.

Dr P. Heil is a Senior Research Scientist with the Australian Antarctic Division working on sea ice physics, including modelling and observations. Dr Heil leads an international team investigating sea ice thickness and volume using numerical modelling, satellite and *in situ* observations. She will contribute 0.15 FTE to this project with a focus on setting up CICE6 in stand-alone mode with Dr Bennetts, and co-leading evaluation of biases within the coupled system with Dr Sandery and Dr Hobbs. Dr R. Massom (sea ice model evaluation, 0.05 FTE), Dr B. Galton-Fenzi (Antarctic coastal circulation, 0.1 FTE) and Dr G. Hyland (0.1 FTE) also comprise AAD technical contributions.

Dr R. Matear is the main developer of the WOMBAT biogeochemical model and leads the Decadal Climate Forecasting project at CSIRO Oceans and Atmosphere. He will contribute 0.1 FTE to the project, leading the BGC modelling. Dr Matear and his team bring a wealth of experience in ocean modelling to the project, and the decadal prediction project is an end-user for COSIMA models.

Dr P. Sandery is a research scientist and developer of atmosphere-ocean-sea-ice analysis and forecasting systems at CSIRO Oceans and Atmosphere. He has extensive experience in the development and verification of Ensemble Kalman Filter data assimilation systems. Dr Sandery and his team will work on the continued integration of ACCESS-OM2/3 into Australia's ocean and sea-ice reanalysis systems (BRAN) and will contribute to improving the model through parameter estimation research efforts that will provide a more rigorous assessment of model error, in particular by co-leading the sea ice bias assessment study (with Dr Heil and Dr Hobbs).

Dr S. Marsland is the leader of the Coupled Modelling Team at CSIRO Oceans and Atmosphere, responsible for delivering the ACCESS-CM2 coupled climate model that is being used for Australia's contribution to the 6th Coupled Model Intercomparison Project. Dr Marsland will contribute 0.1 FTE to this project, and will co-lead the large-scale circulation evaluation in the global models with Prof England.

Dr R. Woodham is the Director of the Maritime Geospatial Program in the Hydrography, Meteorology and Oceanography Branch of the Australian Geospatial-Intelligence Organisation, part of the Department of Defence. Dr Woodham is not a research participant, but his unit will contribute financially to the partnership and has a deep interest in the improvement of Australian ocean modelling for the advancement of Defence capability.

Dr S. Griffies is one of the world's leading ocean modellers, employed at NOAA's GFDL. Dr Griffies will contribute 0.05 FTE to the project, and will form a vital link between COSIMA and the international MOM6 community.

FEASIBILITY AND COMMITMENT

This proposal combines the strength of university researchers and the National Computational Infrastructure (NCI) with the agencies responsible for Australia's Antarctic operations, ocean forecasts, decadal climate prediction and climate projections. It will support a nationwide community of ocean modellers, and will expand the program to bring in new cutting-edge models (including MOM6, CICE6, WOMBAT, WAVEWATCH-III). It will train the next generation of ocean and sea-ice modellers, by attracting talented students to undertake PhD studies in the field. It will culminate in the creation of Australia's next-generation ocean-sea ice model which will represent a transformation in our modelling systems, and maintain Australia's position at the forefront of ocean model developments. The program gains outstanding efficiencies by combining the efforts of multiple institutions, with a strong team with national reach and external engagement, and thereby represents outstanding value for money.

The primary motto of the COSIMA consortium over the last 4 years has been that **the community comes first**. This philosophy has forged our thinking – we have shared model code, output data streams from models and techniques to analyse this output. The result has been a shared vision, across institutions which had previously experienced difficulties in effectively collaborating. The support of this consortium has forged bonds between these institutions, and has led directly to a new proposal with broader scope and greater ambitions than the first COSIMA bid.

The commitment from partners on this project is commensurate with the ambitions. The project has been boosted by direct cash investment from the Department of Defence, through the Hydrography and METOC Branch, which also funds the Bluelink model. Partner organisations have extensive corporate knowledge and experience in ocean-sea ice modelling, and therefore have a strong commitment and interest in the success of the proposal. The plan of work (Fig. 3) has been carefully crafted to take full advantage of the available specialist skills in the consortium (e.g. sea ice models, regional models, surface waves, biogeochemistry) obtaining full value for the outputs of the project.

BENEFIT

The COSIMA project has enormous benefits for Australia. There is an obvious benefit for the industry partners, who will adopt the models and systems that are developed in this project. These industry partners have a mandate to generate operational products, such as ocean state and surface wave forecasts (used in search and rescue, defence operations, shipping, etc.) and decadal climate prediction (of significant benefit to the water resource and agricultural sectors). The proposal will thereby contribute to the Australian Government's National Science and Research Priority 2 (Soil and Water); item 1 (new and integrated national observing systems, technologies and modelling frameworks across the soil-atmosphere-water-marine systems) and 8 (Environmental change); item 1 (Improved accuracy and precision in predicting and measuring the impact of environmental changes caused by climate and local factors).

A significant component of this program is in the modelling of sea ice, and in the near-Antarctic zone. Compared with other components of earth system model development, sea ice modelling is under-resourced at BoM/CSIRO and lacks depth within Australia. This proposal will facilitate critical mass in this area, in particular through funding top-up scholarships for new PhD students to attract the best and brightest to the field. It will also provide a viable capability for modelling the ocean cavities that are found under ice shelves. It will thereby contribute to the current Australian Antarctic Science Strategic Plan (AASSP; 2011/12 to 2020/21) and is well-placed to deliver into the soon-to-be-released AASSP based on current national and international priorities. Furthermore, this proposal is in a prime position to deliver into upcoming AAD (and international) logistics, including the new Australian icebreaker, and possibly year-around aviation for AAD, which require weather forecasts that are strongly influenced by sea ice.

One of the defining aspects of COSIMA over the last 4 years has been to provide all of our products free of charge to the national and international research community. Furthermore, all model and code development is completely open and transparent, hosted on the COSIMA GitHub repository. This approach eliminates barriers to collaboration, and enhances the effectiveness of the investment. It also means that there are no IP limitations in the proposed work.

The proposal is focussed on providing technical advances to the community, but also has direct relevance to scientific research into the ocean-sea ice and climate system. We will provide tools to research students and postdoctoral fellows across the university sector, and we will support the scientific endeavours of the ARC Centre of Excellence for Climate Extremes (of which CIs Hogg and England are Chief Investigators). These research outcomes will include higher-resolution regional studies using MOM6 that better represent Antarctic processes, improved understanding of the dynamics of sea ice, better understanding of the tropical (especially Pacific) ocean, and interactions between the large-scale circulation and fine-scale processes.

COMMUNICATION OF RESULTS

The results of this project will be disseminated as they become available through publication in the top discipline journals such as *J. Phys. Oceanogr.*, *JGR-Oceans*, *Ocean Modelling, Journal of Advances in Modeling Earth*

Systems, and *Geophysical Research Letters*. All new model codes, algorithms, and diagnostic software generated by this project will be documented and published via version-controlled open-source Github repositories (see, for example, github.com/COSIMA). We will also publish all new model simulations that have the potential for broader community use through an expansion of the existing COSIMA model output repository (doi:10.4225/41/5a2dc8543105a).

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